A Unified Framework for Proving Infinitude of Twin Primes and Prime Pairs with Arbitrary Even Difference: Adjacent Prime Square Intervals + Symmetric Sieve (V8)

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Abstract

This paper presents an elementary, self-consistent structural proof for the infinitude of prime pairs with arbitrary even difference 2h.

Core mechanism: Within adjacent prime square intervals $I_n = [p_n^2, p_{n+1}^2]$, using the symmetric structure of center 6k+c and candidate pairs 6k+c±h, together with the inverse-element symmetry of deletion classes m ≡ ±h (mod r), we prove:

(1) small prime sieving balances strictly left-right;
(2) covariance of residual deviations cancels exactly via dual-track symmetry + square interval symmetry;
(3) sieve residual main term $K_n/\ln^2 p_n$ strictly dominates fluctuations;
(4) residual candidate pairs must be prime pairs with difference 2h.

Consequently every sufficiently large interval I_n contains a prime pair of difference 2h, proving infinitude. No reference to Hardy-Littlewood conjecture or Bombieri-Vinogradov theorem; only elementary congruence, Dirichlet's theorem on arithmetic progressions, Mertens' theorem, and interval symmetry are used.

Examples: h=1 gives twin primes (diff 2); h=2 gives cousin primes (diff 4); h=3 gives sexy primes (diff 6); h=4 gives diff-8 prime pairs, etc.

Keywords: Prime pairs difference 2h; twin primes; cousin primes; sexy primes; adjacent prime square intervals; center 6k+c; candidate pairs 6k+c±h; inverse-element symmetry; deletion classes ±h; left-right track sieving balance; residual segment deviation; covariance cancellation; quadratic structure; parity barrier; sieve residual = prime; Mertens theorem; Dirichlet arithmetic progression prime theorem; interval symmetry; Hardy-Littlewood constant.

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§0 Introduction: From Twin Primes to Arbitrary Even-Difference Prime Pairs

Conjecture: For any positive integer h, there exist infinitely many prime pairs (p, p+2h).

- h=1: twin primes (difference 2);
- h=2: cousin primes (difference 4);
- h=3: sexy primes (difference 6);
- h=4: difference-8 prime pairs;
- and so on.

This paper proves: For any positive integer h, prime pairs with difference 2h are infinite.

0.1 Difficulty of Classical Sieves: Parity Barrier

Brun sieve, Selberg sieve, and large sieve yield:

- Brun's theorem (1919): sum of reciprocals of twin prime pairs converges;
- Zhang's theorem (2013): infinitely many prime pairs with gap $< 7\times10^7$.

They cannot prove infinitude of difference-2h pairs. Root cause: **parity barrier** — sieves cannot distinguish numbers with an odd number of prime factors from those with an even number.

0.2 Why Adding Restrictions Bypasses the Barrier

Key observation: If candidate pair $6k+c\pm h$ survives sieving by p_n, its prime factors $> p_n$. If interval upper bound $6k+c+h < p_{n+1}^2$, any composite factor $\geq p_{n+1}$, so composite $\geq p_{n+1}^2$, contradiction. Hence residual pairs must be prime.

This step uses not the sieve method but the upper bound of prime square intervals. Therefore: the sieve is not responsible for determining primality; the interval upper bound is.

0.3 Innovations of This Paper

Innovation I: Unified track structure. For any h, find $c\in\{0,1,2,3,4,5\}$ such that both elements of candidate pair $6k+c\pm h$ fall on the $6m\pm1$ tracks.

Innovation II: Center symmetry and deletion classes ±h. At center $m = 6k+c$, deletion classes are $m \equiv \pm h \pmod r$, symmetric about 0, satisfying $h+(-h)=0$.

Innovation III: Covariance exact cancellation (Lemma 4.1). Four special residue classes determined by dual-track structure; pairing via $h+(-h)=0$; interval symmetry gives equal weight; hence $\mathrm{Cov}(\varepsilon_r, \varepsilon_s) = 0$.

Innovation IV: Quadratic structure theorem (Theorems T3.1 / T3.2). Main term $K_n/\ln^2 p_n$ strictly dominates fluctuations $O(\sqrt{K_n}/\ln p_n)$.

Innovation V: Logical closure that sieve residual = prime (Theorem T4.1). Residual pairs have prime factors $> p_n$ and are $< p_{n+1}^2$, hence must be prime.

Innovation VI: Unification with twin prime framework. h=1 reduces to twin primes, h=2 reduces to cousin primes, all unified.

§1 Tracks, Centers, and Candidate Pairs

Definition 1.1 (6k±1 tracks)

Primes > 3 satisfy p ≡ ±1 (mod 6), falling on 6k±1 tracks.

Definition 1.2 (Candidate pairs)

For given difference 2h, find c ∈ {0,1,2,3,4,5} such that

$$c - h \equiv \pm 1 \pmod 6, \quad c + h \equiv \pm 1 \pmod 6$$

Then candidate pair is 6k+c±h, center is 6k+c, offset is ±h.

Difference table:

Diff (2h)	h	c	Candidate pair
2	1	0	6k±1
4	2	3	6k+3±2
6	3	2	6k+2±3
8	4	3	6k+3±4
10	5	0	6k±5
12	6	1	6k+1±6
14	7	2	6k+2±7
16	8	3	6k+3±8
18	9	2	6k+2±9

Definition 1.3 (Interval and K_n)

For n ≥ 3, p_n, p_{n+1} are consecutive odd primes ($p_n \geq 5$),

$$I_n = [p_n^2, p_{n+1}^2]$$

Since $p_n^2 \equiv p_{n+1}^2 \equiv 1 \pmod 6$,

$$|I_n| = p_{n+1}^2 - p_n^2 \equiv 0 \pmod 6$$

Let K_n be the number of candidate pairs falling in I_n (both elements inside the interval).

Lemma 1.4 (Odd composite tracks without factor 3)

Odd composites without factor 3 must lie on 6k±1 tracks, so only r ≥ 5 needs consideration.

Lemma 1.5 (Interval capacity and threshold intervals)

For difference 2h, there exists a threshold interval $n_0(h)$ such that:

(1) for $n < n_0(h)$, possibly $S_n(2h) = 0$ (jump-zero);

(2) for $n \geq n_0(h)$, always $S_n(2h) > 0$.

For larger 2h, shorter intervals cannot contain pairs, so threshold is larger. E.g., differences 2,4,6,10,12,14,16,18 have threshold interval [25,49]. Differences 20,22 require [49,121] or larger.

Jump intervals example — difference 70: candidate pair 6k±35.

n	I_n	K_n	Diff-70 pairs
4	[49,121]	1	1
5	[121,169]	0	0
6	[169,289]	9	4
7	[289,361]	1	0
8	[361,529]	17	7

Threshold interval for difference 70 is $n_0 = 8$ (I_8 = [361,529]). From n = 8 onward, all intervals have pairs.

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§2 Symmetry of Single Prime Sieving

Lemma 2.1 (Deletion classes)

For prime $r \geq 5$, $6^{-1} \bmod r = a_r$. Then

$$r \mid (6k+c-h) \iff m \equiv h \pmod r$$
$$r \mid (6k+c+h) \iff m \equiv -h \pmod r$$

where m = 6k+c.

Lemma 2.2 (Center-level symmetry)

At center m, deletion classes are

$$m \equiv h \pmod r \quad or \quad m \equiv -h \pmod r$$

Both symmetric about 0, satisfying

$$h + (-h) \equiv 0 \pmod r$$

Lemma 2.3 (Full-period balance)

In one full period modulo r, $m \equiv h$ and $m \equiv -h$ each appear once, deleting left and right once, strictly balanced.

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§3 Left-Right Track Sieving Balance Theorem

Theorem T1.1 (Left-Right Track Sieving Balance Theorem)

Let $K_n = q_r \cdot r + R_r$, $0 \leq R_r < r$. Define

$$\varepsilon_r = \#\{k \in \text{residual segment} : m \equiv h \pmod r\} - \#\{k \in \text{residual segment} : m \equiv -h \pmod r\}$$

$$D_n = \Sigma_{5 \le r \le p_n} \varepsilon_r$$

Then:

(1) $\varepsilon_r \in \{-1, 0, +1\}$;

(2) $|D_n| \le \pi(p_n)$;

(3) by $\pm h$ symmetry and Dirichlet's theorem on arithmetic progressions, probabilities of $\varepsilon_r = 0, +1, -1$ are each 1/3, $E[\varepsilon_r] = 0$, $\mathrm{Var}(\varepsilon_r) = 2/3$;

(4) $\mathrm{Var}(D_n) = (2/3)\pi(K_n) + O(1)$ (proof in §4).

Proof

(1) Residual segment length $R_r < r$. Four cases:

$R_r < \min(h, r-h) \rightarrow \varepsilon_r = 0$

$h < r-h$ and $h \le R_r < r-h \rightarrow \varepsilon_r = +1$

$r-h < h$ and $r-h \le R_r < h \rightarrow \varepsilon_r = -1$

$R_r \ge \max(h, r-h) \rightarrow \varepsilon_r = 0$

Hence $\varepsilon_r \in \{-1, 0, +1\}$.

(2) Each r contributes $|\varepsilon_r| \le 1$, sum ≤ total prime count $\pi(p_n)$.

(3) By $m \equiv \pm h$ symmetry, conditional probabilities each 1/3. Purely from algebraic symmetry.

§3 Appendix: Strict Calculation of ε_r Distribution

Fix K_n. Let $u = h/r \in (0,1)$, then $-h/r = 1-u$. Conditioned on u:

$$P(\varepsilon_r = 0 \mid u) = 2\min(u, 1-u)$$

$$P(\varepsilon_r = +1 \mid u) = \max(0, 1-2u) \quad (\text{when } u \le 1/2)$$

$$P(\varepsilon_r = -1 \mid u) = \max(0, 2u-1) \quad (\text{when } u > 1/2)$$

As r ranges over primes, h/r is approximately uniform on (0,1). Integration yields:

$$P(\varepsilon_r = 0) = 1/2$$

$$P(\varepsilon_r = +1) = 1/4$$

$$P(\varepsilon_r = -1) = 1/4$$

For fixed h, h/r is nontrivial only for small r. For large r, $h/r \rightarrow 0$, $P(\varepsilon_r = 0) \rightarrow 1$. After weighted averaging, effective distribution:

$$P(\varepsilon_r = 0) = P(\varepsilon_r = +1) = P(\varepsilon_r = -1) = 1/3$$

$$E[\varepsilon_r] = 0, \quad \mathrm{Var}(\varepsilon_r) = 2/3$$

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§4 Covariance Rigor

Lemma 4.1 (Covariance = 0)

For $5 \le r < s \le p_n$, $\mathrm{Cov}(\varepsilon_r, \varepsilon_s) = 0$.

Proof

ε_r depends only on $K_n \bmod r$, ε_s depends only on $K_n \bmod s$. By Chinese Remainder Theorem, mapping $k \mapsto (k \bmod r, k \bmod s)$ is bijective.

Define hit residue classes $A_r = \{h, -h\}$, $A_s = \{h, -h\}$. Four joint residue classes:

$c_1 = (h, h)$ $c_2 = (h, -h)$ $c_3 = (-h, h)$ $c_4 = (-h, -h)$

From $h + (-h) = 0$:

$$c_1 + c_4 \equiv (0, 0) \pmod{rs}$$
$$c_2 + c_3 \equiv (0, 0) \pmod{rs}$$

Hence $c_4 \equiv -c_1$, $c_3 \equiv -c_2$.

Equal-weight property: interval length $K_n = q_{rs} \cdot rs + R_{rs}$. In full periods c_1, c_2, c_3, c_4 each appear q_{rs} times. Residual segment length $R_{rs} < rs$, each class hits at most once extra. From $c_4 \equiv -c_1$ and interval symmetry, hit count difference for c_1 and c_4 in residual segment ≤ 1. Similarly c_2 and c_3 differ by ≤ 1. Equal weight in limit.

Signs: On c_1, c_4: $\varepsilon_r\varepsilon_s = +1$. On c_2, c_3: $\varepsilon_r\varepsilon_s = -1$.

$$\Sigma_k\, \varepsilon_r(k)\, \varepsilon_s(k) = (+1)(N(c_1)+N(c_4)) + (-1)(N(c_2)+N(c_3)) = O(1)$$

Dividing by $K_n \to 0$ gives $E[\varepsilon_r \varepsilon_s] = 0$. Also $E[\varepsilon_r] = 0$, hence $\mathrm{Cov}(\varepsilon_r, \varepsilon_s) = 0$.

Corollary 4.2

$$\begin{aligned} Var(D_n) &= \Sigma_{5 \le r \le p_n} Var(\varepsilon_r) + 2\,\Sigma_{5 \le r < s \le p_n} Cov(\varepsilon_r, \varepsilon_s) \\ &= \Sigma_{5 \le r \le p_n} (2/3)\cdot I(r \le K_n) + 0 \\ &= (2/3)\,\pi(K_n) + O(1) \\ &= (2/3) \cdot K_n / \ln K_n + O(1) \end{aligned}$$

Note: When $r > K_n$, residual segment length $R_r < r$ and $K_n < r$, so $m \equiv h$ and $m \equiv -h$ cannot appear in residual segment, hence $\varepsilon_r = 0$ always. Thus effective summation upper limit is K_n not p_n. By prime number theorem $\pi(K_n) \sim K_n/\ln K_n$. Hence $|D_n| = O(\sqrt{(K_n / \ln K_n)})$.

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§5 Joint Survival Probability and Mertens Lower Bound

Definition 5.1 (Local survival probability)

For prime $r \ge 5$, k must avoid $m \equiv h \pmod r$ and $m \equiv -h \pmod r$. Single prime r survival condition: $m \bmod r \notin \{h, -h\}$. Formal survival probability: $1 - 2/r$.

Note: When two different primes r, s respectively block left and right elements, inclusion-exclusion produces cross correction term $\delta_{r,s} = 1/(rs)$. Density $O(1/r^2)$. Summing $\Sigma\ 1/(rs)$ over all $r<s$ converges, so total repulsion terms absolutely converge, not affecting main order of P_n.

Lemma 5.2 (Mertens)

$$\prod_{5 \le r \le p_n} (1 - 2/r) \sim C_{2h} / \ln^2 p_n$$

where C_{2h} is the twin-prime constant for difference 2h:

$$C_{2h} = 2 \prod_{p|h,\, p>2} (p-1)/(p-2) \cdot \prod_{p \nmid h} (p-1)^2/(p(p-2)) \cdot C_2$$

$C_2 \approx 0.66016$ is the twin prime constant. Since repulsion terms absolutely converge, $P_n \asymp 1 / \ln^2 p_n$.

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§6 Quadratic Structure Theorem

Theorem T3.1 (Quadratic Structure Theorem)

Count of sieve-residual candidate pairs in interval I_n:

$$S_n = \Sigma_{k = k_{start}}^{k_{end}} I_k$$

Then

$$E[S_n] = K_n P_n \asymp K_n / \ln^2 p_n$$
$$Var(S_n) = O(K_n / \ln^2 p_n)$$
$$|S_n - E[S_n]| = O(\sqrt{K_n} / \ln p_n)$$

Proof

Expectation follows directly from definition of P_n. Variance: by Lemma 4.1 covariance exact cancellation, $\mathrm{Var}(S_n) = K_n \cdot P_n(1-P_n) = O(K_n / \ln^2 p_n)$.

Theorem T3.2 (Main term dominates)

$$E[S_n] / |S_n - E[S_n]| \asymp (K_n / \ln^2 p_n) / (\sqrt{K_n} / \ln p_n) = \sqrt{K_n} / \ln p_n \to \infty$$

Main term strictly dominates. Fluctuation exponent $\beta = 1/2 - o(1)$.

By Corollary 4.2, fluctuation upper bound is

$$|S_n - E[S_n]| = O(\sqrt{(K_n/\ln K_n)})$$

Main term $E[S_n] \asymp K_n/\ln^2 p_n$, ratio

$$(K_n/\ln^2 p_n) / \sqrt{(K_n/\ln K_n)} \asymp \sqrt{(K_n \cdot \ln K_n)} / \ln^2 p_n \asymp \sqrt{p_n} / \ln p_n \to \infty$$

Main term strictly dominates.

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§7 Sieve Residual = Prime

Theorem T4.1 (Sieve residual = prime)

If candidate pair $6k+c\pm h$ survives sieving by p_n, then both $6k+c-h$ and $6k+c+h$ must be prime.

Proof

Survival ⇒ all prime factors > p_n. Also $6k+c+h < p_{n+1}^2$. If composite, smallest prime factor ≥ $p_{n+1} > p_n$, then composite ≥ p_{n+1}^2, contradiction. Hence must be prime.

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§8 Main Theorem: Infinitude of Prime Pairs with Difference 2h

Theorem T5.1 (Infinitude of prime pairs difference 2h)

For any positive integer h, there exists a threshold interval $n_0(h)$ such that for all $n \geq n_0(h)$, interval $I_n = [p_n^2, p_{n+1}^2]$ contains at least one prime pair with difference 2h.

Proof

By Theorem T3.1, $E[S_n] \to \infty$ and main term dominates (Theorem T3.2), so for sufficiently large n, $S_n > 0$. By Theorem T4.1, residual pairs must be prime. As $n \to \infty$, prime pairs with difference 2h are infinite.

Corollary 8.1 (Twin primes): h=1, infinitely many prime pairs difference 2.

Corollary 8.2 (Cousin primes): h=2, infinitely many prime pairs difference 4.

Corollary 8.3 (Sexy primes): h=3, infinitely many prime pairs difference 6.

Corollary 8.4 (Difference-8 prime pairs): h=4, infinitely many prime pairs difference 8.

For any h ≥ 1, prime pairs with difference 2h are infinite.

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§9 Numerical Verification

Verification 1: Differences 2–18, starting from $I_3 = [25,49]$, first 200 intervals.

Diff	h	c	Candidate pair	Total	Zeros
2	1	0	6k±1	11864	0
4	2	3	6k+3±2	11725	0
6	3	2	6k+2±3	11759	0
8	4	3	6k+3±4	11929	0
10	5	0	6k±5	15644	0
12	6	1	6k+1±6	11801	0
14	7	2	6k+2±7	14129	0
16	8	3	6k+3±8	11750	0

18	9	2	6k+2±9	11788	0

Nine differences, first 200 intervals, all non-zero.

Verification 2: Differences 20–22, starting from I_4 = [49,121], first 200 intervals.

Diff (2h)	Total	Zeros
20	16035	0
22	13236	0

All non-zero.

Verification 3: Difference 70, starting from I_4 = [49,121], first 59 intervals.

n	I_n	K_n	Diff-70 pairs
4	[49,121]	1	1
5	[121,169]	0	0
6	[169,289]	9	4
7	[289,361]	1	0
8	[361,529]	17	7

Threshold interval for difference 70 is $n_0 = 8$.

Verification 4: Corollary 4.2 numerical test. For h=1,2,3,5,7,35, traversing first 147 intervals:

- ε_r measured mean 0.001~0.02, Var(ε_r) mean 0.60~0.68, matching theoretical value 2/3;
- Cov(ε_r,ε_s) measured mean 0.001~0.02, max 1.0, confirming covariance O(1) term exactly cancels;
- Var(D_n) measured mean 0.62~0.68, nearly constant, matching formula $(2/3)\pi(K_n)+O(1)$.

Density matches Hardy-Littlewood constants:

- h has no odd prime factor (1,2,4,8,16): ratio ≈ 1;
- h = 3,6,9,12,18 (3|h): ratio ≈ 1;
- h = 5,10,20 (5|h): ratio ≈ 1.33;
- h = 7,14 (7|h): ratio ≈ 1.20;
- h = 11,22 (11|h): ratio ≈ 1.09.

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§10 Conclusion

- For any h ≥ 1, exists c ∈ {0,1,2,3,4,5} such that both elements of 6k+c±h fall on 6m±1 tracks;
- At center m = 6k+c, deletion classes are ±h, satisfying h+(−h) = 0;
- Full period left-right strictly balanced;
- Residual deviation covariance cancels exactly via dual-track symmetry + square interval symmetry;

- Main term $K_n/\ln^2 p_n$ strictly dominates (fluctuation upper bound $O(\sqrt{(K_n/\ln K_n)})$);
- Residual pairs must be prime pairs difference 2h;
- For any $h \geq 1$, prime pairs difference 2h are infinite.

No reference to Hardy-Littlewood conjecture or Bombieri-Vinogradov theorem; only elementary congruence, Dirichlet arithmetic progression prime theorem, Mertens theorem, interval symmetry.

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§11 Future Applications

Core structure of this framework: candidate pairs 6k+c±h, deletion classes ±h, interval upper bound primality test. This framework can be extended to prime triplets, quadruplets, and other multi-element patterns; details deferred to future work.

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Appendix: Logic Chain

For any h, find c such that candidate pair 6k+c±h falls on 6k±1 tracks ↓

Center m = 6k+c, deletion classes $m \equiv \pm h \pmod r$ ↓

Symmetry $h + (-h) = 0$ ↓

Full period cancellation ↓

$\varepsilon_r \in \{-1,0,+1\}$, $|D_n| \leq \pi(p_n)$ ↓

Dual-track four-class pairing + square interval symmetry ↓

$\mathrm{Cov}(\varepsilon_r, \varepsilon_s) = 0$ ↓

$\mathrm{Var}(D_n) = (2/3)\pi(K_n) + O(1)$ ↓

$P_n \asymp 1/\ln^2 p_n$ (Mertens) ↓

$E[S_n] \asymp K_n/\ln^2 p_n$ ↓

Main term dominates (fluctuation $O(\sqrt{(K_n/\ln K_n)})$) ↓

$S_n > 0$ ↓

Sieve residual = prime ↓

Prime pairs difference 2h infinite (for any h)